Josh Seidman, PhD Candidate

Marquette University
Klinger College of Arts & Sciences
Department of Mathematical and
Statistical Sciences

West Allis, WI, 53214
Email: josh.seidman@marquette.edu
GitHub: https://github.com/SideofMan
Phone: 630-408-3680

Professional Summary

Applied mathematician eager to launch a career in scientific machine learning research. Experienced in machine learning, uncertainty quantification, and data analysis. Highly adaptable and has the proven ability to solve complex problems using Python, R, and SQL and clearly communicate findings and solutions to audiences without technical backgrounds.

Technical Skills

- Programming: R, Python, SQL, C++, MATLAB, Mathematica, KQL
- Statistical Analysis Tools: RStudio, Excel, Microsoft Fabric, Microsoft Power BI
- Miscellaneous: Git, Linux, HPC, Microsoft Office

Education

2022-present **Ph.D. in Computational Sciences**, *Marquette University*, Milwaukee, WI (Expected 5/2027), *Advisor: Elaine Spiller*
2022-2024 **M.S. in Applied Statistics**, *Marquette University,* Milwaukee, WI
2019-2021 **M.A. in Mathematics**, *Indiana University*, Bloomington, IN
2016-2019 **B.S. in Mathematics**, *Indiana University*, Bloomington, IN
2016-2019 **B.S. in Physics**, *Indiana University*, Bloomington, IN

Research

Research Assistantship (5/2023 – present)
Marquette University, Department of Mathematical and Statistical Sciences, Milwaukee, WI

- Developed application of Vecchia approximation to the parallel partial emulator (VPPE).
  - Developed packages for implementation in R (https://github.com/SideofMan/VPPE).
  - Paper has been published in JUQ (https://epubs.siam.org/doi/10.1137/25M1789822).
  - Presented at SIAM's Uncertainty Quantification conference, Minneapolis, 2026.
  - Presented at the Symposium on Data Science and Statistics, Milwaukee, 2026.
- Developing an AI debris flow model to adaptively predict entrainment rates.
  - Utilizes local topography and flow parameters to train the neural network to predict entrainment rates depending on the location and flow parameters.
- Wrote R scripts that implement two algorithms.
  - Imputation for a zero-censored GP (https://github.com/SideofMan/zGP).
  - Adaptive annealed importance sampling for evaluating marginal likelihoods in Bayesian model comparison (https://github.com/SideofMan/AAIS_R).
  - Presented at SIAM's Uncertainty Quantification conference, Trieste, Italy, 2024.
- Implemented gradient kernel dimension reduction for a parallel partial linked emulator.
- Developed the grouped deep Gaussian process to efficiently emulate geophysical processes with large output dimensions.

- Implementation of custom kernels to parallel partial emulation for improved prediction on models with low number of training runs.
- Working with the College of Nursing to develop a clinical prediction tool for screening substance abuse-related emergency department visits using the National Emergency Department Sample dataset.

Leadership

Student Success Seminar Organizer (8/2025 – present)
Marquette University, Department of Mathematical and Statistical Sciences, Milwaukee, WI

- Organized seminar for graduate students in the department to give math talks.
- Coordinated speakers biweekly via sign-up sheet and email invitations.
- Managed scheduling, promotion of the seminar, and refreshments.

Graduate Student Organization Department Representative (8/2024 – present)
Marquette University, Department of Mathematical and Statistical Sciences, Milwaukee, WI

- Represented math department graduate students before GSO leadership, advocating for their concerns and interests.
- Organized semesterly department town halls to gather feedback on how the GSO can help improve grad student life at Marquette.

Discovery World Dream Machine Hackathon Co-Lead (2/2026)
University of Wisconsin-Milwaukee, Connected Systems Institute, Milwaukee, WI

- Helped lead team to win the grand prize for best overall solution.
- Delegated tasks for designing and the dashboard.

Publications

Peer-Reviewed Journal Articles – Published

- J. Seidman, E.T. Spiller. *VPPE: Application of Scaled Vecchia Approximations to Parallel Partial Emulation*. SIAM/ASA J. Uncert. Quant, https://doi.org/10.1137/25M1789822.

Peer-Reviewed Journal Articles – In Preparation

- J. Seidman, E.T. Spiller. *Grouped Deep Gaussian Process Emulation using Stochastic Imputation for handling non-stationarity and massive model output.*

Short-form Peer-Reviewed Conference Papers – Published:

- J. Seidman, E.T. Spiller. *A Robust Scaled Vecchia Gaussian Process Approach for Computer Model-Based Learning of Physical Processes*. Symposium on Data Science & Statistics 2026.

Honors/Awards

- **Earl W. Swokowski Fellowship**, *College of Arts and Sciences, Marquette University*, Fall 2026, Summer 2026.
- **Discovery World Dream Machine Hackathon Grand Prize Winner**, *Connected Systems Institute, University of Wisconsin-Milwaukee,* 2026.
  - Designed a data dashboard for Discovery World's Dream Machine.
  - Used SQL, Power BI, and Microsoft Fabric to analyze data and implement predictive models in the customer facing dashboard displaying Dream Machine data.
  - Won grand prize for best overall solution out of 10 teams.
- **Student Travel Award for the 2026 SIAM Conference on Uncertainty Quantification**, *National Science Foundation*, 2026.
- **Symposium on Data Science & Statistics 2026 Scholarship Fund Recipient**, *Northwestern Mutual Data Science Institute,* 2026.

- **S.J. Merrill Award for Best Teaching Assistant**, *Department of Mathematical and Statistical Sciences, Marquette University*, 2025.
- **David A. Rothrock Teaching Fellowship: Excellence of Teaching of Mathematics**, *College of Arts & Sciences, Indiana University*, 2020.
- **Thelma Abell Prize: High Scholastic Merit in Mathematics**, *College of Arts & Sciences, Indiana University*, 2019.
- **Gabriel Benman Memorial Award: Outstanding Promise and Achievement in Physics**, *College of Arts & Sciences, Indiana University*, 2019.
- **Mathematics Hutton Honors Dean Scholarship**, *College of Arts & Sciences, Indiana University*, 2016-2019.

Teaching

Graduate Teaching Assistant (8/2022 – present)
Marquette University, Department of Mathematical and Statistical Sciences, Milwaukee, WI
- Instructor of Record for Precalculus (Fall 2026)
- Instructor of Record for Calculus for the Biological Sciences (Spring 2026)
- Elementary Statistics (Spring 2025)
- Differential Equations (Fall 2023)
- Calculus 2 for Biomedical and Civil Engineers (Spring 2023, Spring 2024, Fall 2024)
- Calculus 2 (Fall 2022)

High School Mathematics Teacher (8/2021 – 5/2022)
Lake Zurich Community USD 95, Department of Mathematics, Lake Zurich, IL
- Taught Algebra 2 and Geometry, including basic statistics and modeling.
- Collaborated with other teachers to utilize data collection to improve curriculum.

Graduate Associate Instructor (8/2019 – 5/2021)
Indiana University, Department of Mathematics, Bloomington, IN
- Finite Mathematics (Spring 2021)
- Introduction to Algebra (Fall 2020)
- College Algebra (Fall 2019, Spring 2020)

Peer Assisted Study Sessions Graduate Student Assistant (1/2020 – 5/2021)
Indiana University, Student Academic Center, Bloomington, IN
- Ran the Peer Assisted Study Sessions program at Indiana University.
- Managed undergraduate tutors and improved student performance in math courses.

Memberships

American Mathematical Society
Honor Society of Phi Kappa Phi
Society for Industrial and Applied Mathematics